Candidate Seat no………….

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech (CE) Examination

November 2015

CE313 System Software

Date: 28.11.2015, Saturday **Time: 10.00 a.m. To 01.00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises two sections and both must be attempted in separate answer sheets.
2. Make suitable assumptions and draw neat figures wherever required.
3. Write entire option in case of multiple choice and matching questions.

SECTION – I

Q – 1 Do as directed.

(A) Choose correct or the best alternative in the following. **[03]**

1) Which of the following loader is executed when a system is first turned on or restarted?
   (a) Boot loader (b) Bootstrap loader
   (c) Compile and Go loader (d) Relating loader
2) Which one of the following will not accept the string 1001?
   (a) 10*1 (b) (00|01|10|11)*
   (c) 1*01* (d) (0|1)*
3) The translator which perform macro expansion is called a
   (a) Macro Processor (b) Macro Pre-processor
   (c) Micro Processor (d) Micro Pre-processor

(B) Match the following. **[05]**

1) Round Rabin Scheduling	a) Analysis phase
2) Semantic rules	b) Macro
3) YACC	c) Time sharing system
4) #define	d) Synthesis phase
5) Code optimization	e) Language processor development tool
	f) Batch processing system

(C) Define left factored grammar. Is following grammar contain Left Factoring? If yes then rewrite the grammar after removing Left Factoring from it. **[03]**

A → cd | cfg | Bi | Bij

B → xyz | z

Q - 2 Answer the following questions. (Any three) **[12]**

(A) What do you mean by a ‘Two-Pass assembler’? Describe working principle of a two-pass assembler.

(B) What is nested macro? Give relevant example. Explain how the nested macro call is processed.

(C) Explain following terms with Example.
1. Keyword Parameter
2. Positional Parameter
3. Default Parameter
4. Mixed Parameter

(D) What do you mean by macro, macro expansion, macro definition and macro call?

(E) What is editor? List types of editor and also mentions the functions of an editor.

Q – 3 Answer the following questions.

(A) Write brief note on resource allocation. Also list and describe popular strategies of it. **[03]**

(B) Define finite state automata. **[04]**

(C) With the help of example list & explain all data structure of single-pass assembler. **[05]**

OR

Q – 3 Answer the following questions.

(A) Distinguish a literal from a constant and Immediate Operand. **[03]**

(B) Define job and batch. Also explain batch processing with functions of batch monitor. **[04]**

(C) Draw a DFA for b a (a | b) * b a **[05]**

SECTION – II

Q - 4 Do as directed.

(A) Answer the following questions. **[05]**

1) Language processors are used to bridge ________ gap.
2) What is difference between START and ORIGIN?
3) Define program.
4) Define semantic expansion with respect to macro.
5) OS start its processing at T1 and stop at time T2 after processing a collection of programs containing N programs this write the equation of system throughput.

(B) Answer the following questions. **[06]**

1) List different flavors of Language processor.
2) Why are we using LTORG statement? What happens if we have not used LTORG statement in our program?
3) Difference between open subroutine and close subroutine.

Q - 5 Answer the following questions. (Any three) **[12]**

(A) What are the functions performed by two Pass Linker?

(B) What do you mean by program relocation? What so you mean by self-relocatable program?

(C) Write a short not on java interpreter.

(D) Differentiate compiler and interpreter.

(E) What do you mean by memory fragmentation? Explain two type of it.

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Q - 6 Answer the following questions.

(A) What are the methods available for debugging in Turbo C++? **[05]**

(B) Process the string A=B*C-30 from each phase of compiler and show the output of each phase. **[05]**

(C) Define alphabet and language. **[02]**

OR

(A) For the given program, do the following. **[05]**

```
        START    200
        MOVER    AREG , ='5'
        MOVEM    AREG , A
LOOP    MOVER    AREG , A
        MOVER    CREG , B
        ADD      CREG , ='1'
NEXT    SUB      AREG , ='1'
        BC       LT , BACK
LAST    STOP
        ORIGIN   LOOP + 2
        MULT     CREG , B
        ORIGIN   LAST + 1
A       DS       1
BACK    EQU      LOOP
B       DS       1
        END
```

1. Show the contents of Symbol table at the end of PASS-1.
2. Show the content of Opcode Table at the end of PASS-1.
3. Show the content of Literal Table and Pool Table.

(B) Derive string id + id * id from given grammar using left and right most derivation. **[05]**

E→ E + E | E * E | id

(C) What is Dynamic Linking? **[02]**

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